Battery System Pitch Deck

**Powering Reliable &**
**Sustainable Energy Systems**

# Advanced Energy Storage Solutions

AVESTA

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Ninove BE

# Who Are We ?

2019 - AVESTA Battery and Energy Engineering Founded by
Prof. Dr. Noshin Omar with a focus on advanced battery research and engineering

Since 2021 – ABEE has expanded within Belgium as outside of Belgium

2024 – Rebranding to AVESTA Holding

Mission - At AVESTA, we are dedicated to designing, manufacturing, and recycling high-performance batteries with a focus on sustainability.
We offer advanced solutions for industries seeking reliable and eco-friendly energy storage, 100% made in Europe.

Vision - AVESTA envisions a world powered by sustainable & efficient energy.
Through continuous innovation in the battery life cycle, we are committed to leading the way toward a cleaner, greener future for all applications and industries.

***Delivering power, jobs, and energy security for generations to come.***

**Reliable Technology**

## 100+ Engineers Globaly

Built on proven, high-quality battery technologies to ensure consistent performance and dependable energy supply in all applications.

**Cost Efficiency**

## 30% to 50% ROI

Optimized solutions designed to reduce energy costs, improve return on investment, and maximize long-term savings.

**Scalable Systems**

## 30 GWh+ Capacity Golobaly

Flexible and modular designs that can be easily expanded to meet growing energy demands across residential, commercial, and industrial sectors.

**Long Lifecycle**

## 8,000 – 12000 cycles

Engineered for durability with extended battery life, reducing replacement frequency and total cost of ownership.

**Technical Support**

## 25+ technician

Dedicated expert support providing guidance, maintenance, and system optimization throughout the entire lifecycle.

**End-of-Life Support**

## 30,000+ Ton recycle capacity

Environmentally responsible recycling and disposal solutions in compliance with Egyptian regulations, promoting sustainability and circular economy practices.

# Our Solutions

We deliver advanced energy storage and battery solutions designed to meet the evolving needs of residential, commercial, industrial, mobility, and utility-scale applications. Our portfolio combines cutting-edge technology with reliability and efficiency to support sustainable energy use, optimize performance, and ensure uninterrupted power supply across all sectors.

All our solutions are developed and manufactured in compliance with European Union standards and certifications, ensuring the highest levels of safety, quality, environmental responsibility, and performance.

## Section 1: Core Technology

**1. SSB Battery** — Next-gen solid-state batteries with higher energy density, improved safety, and longer lifespan. We support R&D, materials sourcing, manufacturing scale-up, and integration guidance.

**2. BMS (Passive & Active)** — Systems that monitor, balance, and protect battery cells. Passive uses resistors to dissipate excess energy; active redistributes energy between cells. We design and optimize BMS architectures and enhance safety protocols.

**3. Recycling** — Recovery of valuable materials like lithium, cobalt, and nickel from end-of-life batteries. We develop recycling processes, improve material recovery rates, and ensure regulatory compliance.

## Section 2: E-Mobility

1. Golf Carts, Forklifts, Scooters — Light electric vehicles and industrial equipment powered by batteries. We provide battery pack design, BMS integration, performance optimization, and safety testing.

2. Railway and Marine — Battery-electric or hybrid propulsion systems for trains, ferries, and ships. We support high-capacity energy storage design, thermal management, certification support, and system integration for heavy-duty cycles.

## Section 3: Stationary Application

BESS for Utility and C&I — Battery Energy Storage Systems for grid-scale (utility) and commercial & industrial use, enabling peak shaving, load shifting, backup power, and renewable integration. We assist with system sizing, technology selection, grid connection, economic modeling, and deployment strategy.

# AVESTA

# POWER YOUR HOME SECURE YOUR FUTURE

Discover next-generation home energy storage solutions designed to deliver energy independence, efficiency, and long-term reliability—engineered in Europe for modern energy needs.

## PRODUCT FEATURES

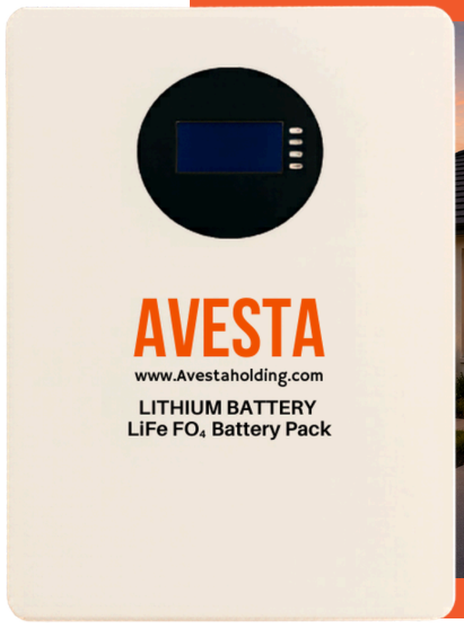
- Advanced lithium-ion battery technology
- High efficiency with extended lifecycle performance
- Intelligent energy management system
- Compact, modern, and scalable design
- Flexible configurations for various residential needs
- Developed according to EU safety, quality, and certification standards

## ABOUT AVESTA

At Avesta Holding, we go beyond delivering advanced battery energy storage systems—we build a fully integrated, future-ready energy ecosystem.

Our solutions are designed, engineered, and validated according to strict EU standards and certifications, ensuring the highest levels of safety, performance, and regulatory compliance.

With a strong foundation in research, manufacturing, and recycling, Avesta Holding plays a key role in supporting the global transition toward sustainable and decentralized energy systems.



### 5 kWh

- ✔ Scalable
- ✔ Solar compatible
- ✔ Easy installation

**Small homes & apartments**

### 16 kWh

- ✔ Scalable
- ✔ Solar compatible
- ✔ Easy installation

**Standard homes**

### 32 kWh

- ✔ Scalable
- ✔ Solar compatible
- ✔ Easy installation

**Large homes & small business**

AVESTA

# GOLF CART BATTERY

## Why Us ?

- European-based Technology
- Full Egyptian supply chain
- customized and optimized modules
- Complete lifecycle support
- Scalable production capacity
- competitive after-sales service
- all products are 100% recyclable
- Warranty up to 10 years

**Built In :**

- Safe, stable battery system with long service life
- Independent charging and discharging protection to prevent over-charging and over-discharging
- Operating temperature: –20 °C to +55 °C
- Integrated On/Off button
- LED indicators for SOC (State of Charge), Run, and Alarm
- Real-time backend data monitoring; remote software (firmware) updates; communication-based system management

**ADD ON's:**

- IP65, suitable for outdoor applications
- LCD display for system information
- Remote control switch
- Bluetooth communication module
- Battery heater for low-temperature environments
- GPS satellite positioning module

| Model | Voltage(V) | Capacity(Ah) | Dimension (mm) | | |
|---|---|---|---|---|---|
| | | | L | W | H |
| 38.4V100Ah | 38.4 | 100 | 385 | 338 | 245 |
| 51.2V100Ah | 51.2 | 100 | 490 | 338 | 245 |
| 51.2V105Ah | 51.2 | 105 | 540 | 330 | 225 |
| 51.2V206Ah | 51.2 | 206 | 1040 | 185 | 230 |
| 73.6V20Ah | 73.6 | 20 | 320 | 200 | 180 |
| 73.6V30Ah | 73.6 | 30 | 360 | 230 | 170 |
| 76.8V100Ah | 76.8 | 100 | 740 | 338 | 245 |
| 76.8V105Ah | 76.8 | 105 | 800 | 360 | 175 |

STORED GREEN, WORK CLEAN, MOVE LEAN

PLUG AND PLAY

AVESTA

# LFP FORKLIFTS BATTERY

## Why Avesta?

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- Full European supply chain
- Scalable production capacity
- Complete lifecycle support
- Customized and optimized modules
- Competitive after-sales service
- All products are 100% recyclable
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- Safe, stable battery system with long service life
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- Real-time backend data monitoring; remote software (firmware) updates; communication-based system management

**ADD ON's:**

- IP65, suitable for outdoor applications
- LCD display for system information
- Remote control switch
- Bluetooth communication module
- Battery heater for low-temperature environments
- GPS satellite positioning module

| Voltage | Series Number | Capacity (Ah) | Dimension (mm) | Weight |
|---|---|---|---|---|
| 24 | B-LFP 24-100 | 100 | 624x207x627 | 212 |
| | B-LFP 24-230 | 230 | 624x284x627 | 288 |
| | B-LFP 24-300 | 300 | 624x284x627 | 288 |
| | B-LFP 24-560 | 560 | 624x428x627 | 435 |
| | B-LFP 24-675 | 675 | 965x343x777 | 727 |
| 36 | B-LFP 36-300 | 300 | 965x345x777 | 744 |
| | B-LFP 36-460 | 460 | 965x337x775 | 748 |
| | B-LFP 36-560 | 560 | 978x461x772 | 1178 |
| | B-LFP 36-615 | 615 | 978x696x575 | 908 |
| | B-LFP 36-690 | 690 | 973x512x775 | 1178 |
| 48 | B-LFP 48-300 | 300 | 830x522x627 | 708 |
| | B-LFP 48-460 | 460 | 830x522x627 | 708 |
| | B-LFP 48-540 | 540 | 830x630x627 | 856 |
| 80 | B-LFP 80-280 | 280 | 1026x564x627 | 965 |

**STORED GREEN, WORK CLEAN, MOVE LEAN**

**PLUG AND PLAY**

AVESTA

STORED GREEN, WORK CLEAN, MOVE LEAN

# ELECTRIC SCOOTER BATTERY

EXPERIENCE SMART, EFFICIENT, AND ECO-FRIENDLY URBAN MOBILITY WITH OUR ELECTRIC SCOOTER. DESIGNED FOR EVERYDAY CONVENIENCE, IT DELIVERS SMOOTH PERFORMANCE, RELIABLE BATTERY EFFICIENCY, MODERN STYLING, AND A COMFORTABLE RIDING EXPERIENCE—MAKING IT THE IDEAL SOLUTION FOR PERSONAL TRANSPORTATION IN TODAY'S FAST-MOVING WORLD.

## Why Us ?

- European-based Technology
- Full Egyptian supply chain
- customized and optimized modules
- Complete lifecycle support
- Scalable production capacity
- competitive after-sales service
- all products are 100% recyclable
- Warranty up to 10 years

AVESTA

| | 261kWh+125kW | 522kWh+250kW |
|---|---|---|
| **Model** | AH-ESS314-C0261 | AH-ESS314-C0522 |
| **Battery** | | |
| Battery Type | LiFePO4 Battery | |
| Rated Energy | 261kWh（832V314Ah） | 522kWh（832V628Ah） |
| **AC(On-grid)** | | |
| Rated Power | 125kW | 250kW |
| AC Voltage | 400Vac/230Vac(L1/L2/L3/N/PE) | 400Vac/230Vac(L1/L2/L3/N/PE) |
| Rated Frequency | 50Hz/60Hz | 50Hz/60Hz |
| Max. AC Current | 189A | 375A |
| **AC(Off-grid)** | | |
| Rated Power | 125kW | 250kW |
| AC Voltage | 400Vac/230Vac(L1/L2/L3/N/PE) | 400Vac/230Vac(L1/L2/L3/N/PE) |
| Rated Frequency | 50Hz/60Hz | 50Hz/60Hz |
| Max. AC Current | 189A | 375A |
| PCS Cooling Method | Intelligent Cooling | |
| **General Parameter** | | |
| Communication Type | RS485/Ethernet/4G | |
| IP Grade | IP54 | |
| Fire Protection System | Yes | |
| Operating Temperature | -10℃~50℃ | |
| Humidity Range | 5%~95% | |
| Altitude | <3000m | |
| Battery Cooling Method | Air-Conditioner | |
| Weight(kg) | 3700kg(about) | 5500kg(about) |
| Size(mm) | 2500*2200*2200mm | 2500*2200*2200mm |

*THE PARAMETERS ARE FOR REFERENCE ONLY, THE ACTUAL PRODUCT SHALL PREVAIL.SUPPORT CUSTOMIZATION, AND ON-DEMAND CUSTOMIZATION

www.bmser.com

BESS:Battery Energy Storage System

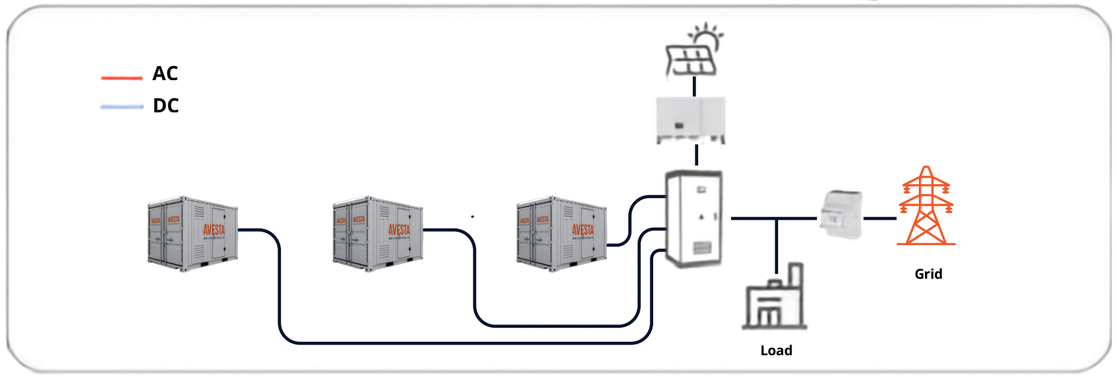
# E-Generator

# 522kWh/2.5MWh

- LiFePO4 Battery+BMS/PCS/EMS
- Easy installation and maintenance
- Supports multiple units in parallel
- Integrated fire protection and air-conditioner

## Why Us ?

- **European-based Technology**
- **Full Egyptian supply chain**
- **customized and optimized modules**
- **Complete lifecycle support**
- **Scalable production capacity**
- **competitive after-sales service**
- **all products are 100% recyclable**
- **Warranty up to 10 years**



# Use Case

## Integration of PV Systems and BESS Into Factories

Real projects. Real capacity. Real returns. ( EU Demonstrations )

| Project | Location | Solar | Battery | Application | Status |
|---|---|---|---|---|---|
| Ninove | Belgium | 660 kW | 1 MWh | Integration of PV Systems and BESS Into Factories | Operational |
| BE Volt Battery Systems | Belgium | 2 MW | 30 MWh | Integration of PV Systems and BESS Into Factories | Under Construction |
| Zeebrugge | Belgium | 8 MW | 50 MWh | Integration of PV Systems and BESS Into Factories | Operational |

### TOTAL DEPLOYED CAPACITY

**10.66 MW**
Solar paired

**81 MWh**
Battery storage

**50,000+**
Operational hours

**1**
Countries active

### WHAT THESE PROJECTS PROVE

- ✓ PV Systems and BESS Into Factories — Zeebrugge 50 MWh handles port peak loads + renewable intermittency
- ✓ C&I cost savings — Ninove 1 MWh cuts peak demand charges
- ✓ PV Systems and BESS Into Factories — BE Volt 30 MWh earns capacity payments + energy arbitrage
- ✓ European validation — EU grid codes, CE certification, real-world performance data
- ✓ Technology maturity — LFP chemistry, active BMS, thermal management proven across climates

### THE INVESTOR ANGLE

"These aren't pilot projects. They're revenue-generating assets with 2–4 years of operational data. The same technology, same team, same manufacturing line — now scaling to Egypt's 14.3 GWh market."

# C&I Industrial – Ninove, Belgium

## 660 kW Solar + 1 MWh BESS Hybrid System

**The Problem:** A 2-hour outage at a cold storage facility destroys €200K in pharmaceuticals. Peak demand charges can make up 40% of an industrial electricity bill. In Belgium, grid instability and rising peak tariffs strain industrial operations year-round.

**The Scenario:** A pharmaceutical distributor in AVESTA's headquarters in Ninove, Belgium installs AVESTA's 660 kW solar + 1 MWh BESS hybrid system with smart peak-shaving software — **this system deployed and proven at AVESTA's own facilities.**

**Spoilage & downtime**

**Peak demand chargereduction**

**Total annual savings**

**Low System cost**

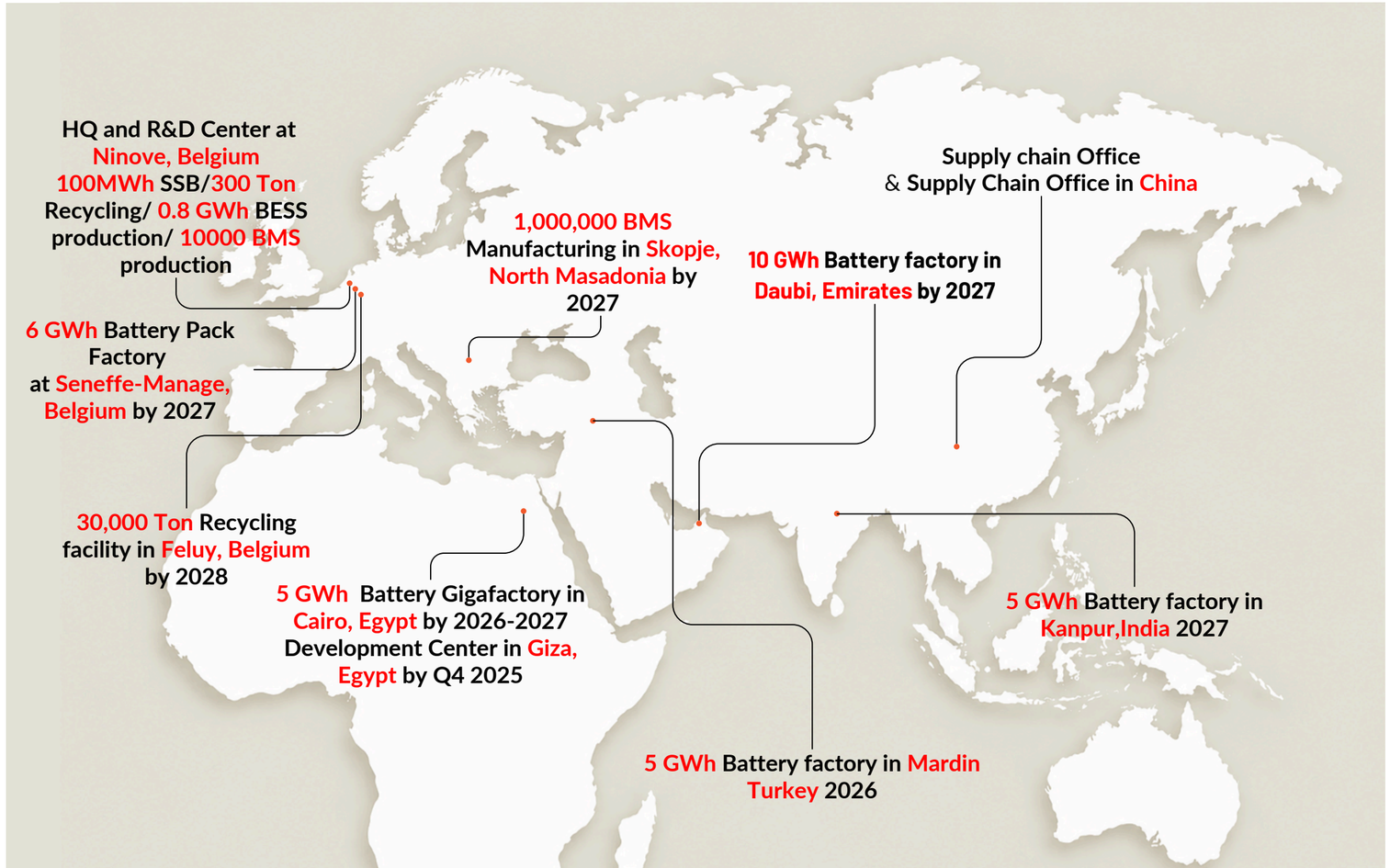
**High ROI payback**

**Solar PV: 660 kWp rooftop (high-efficiency crystalline silicon) | Battery: 1 MWh lithium-ion modular rack-based (containerized) | Power Conversion: Grid-tied PCS with bidirectional inverter**
**Warranty: 5-10 years / 8,000 to 10,000 cycles – backed by AVESTA's local service team. Annual $CO_2$ avoided: 420 tonnes. Self-consumption rate: 85%.**

# Strategic Location

### Community Impact Radius

A strategically positioned global network driving battery production, innovation, and recycling across Europe, the Middle East, and Asia. With 30 GWh capacity, 70K tons recycling, and 1M BMS units, we deliver sustainable growth, support local economies, and accelerate the transition to clean energy.




battery production capacity

## 30+ GWh/ yearly

BMS Manufacturing annually

## 1 million BMS

recycling capacity

## 70,000+ tons

# Three Pillars. One Ecosystem. Built in Egypt.

Kader Factory x AVESTA Holding x Giza Group

### KADER FACTORY

*The Manufacturer*

Kader Factory brings decades of Egyptian industrial manufacturing expertise. Precision engineering. Scale production. Local supply chain mastery. They build the hardware that houses our cells.

**What they deliver:** Fabrication, assembly, quality control, logistics.

### AVESTA HOLDING

*The Technology*

The brain of the operation. Advanced LFP battery research. Cell design. Battery Management Systems. Thermal engineering. 8,000-12,000 cycle technology. 100+ engineers. 30 GWh global network. 5 GWh committed to Egypt.

**What they deliver:** Cells, BMS, software, warranties, recycling.

### G SQUAD

*The Integration*

G Squad Construction is a first-class electrical contracting company, in addition to integrated building works, roads, and corridors. It is one of the sister companies of the Giza Group of Companies (Giza Cables Industry, Giza Cable Accessories, and Giza Ready-Mix Concrete),

**What they deliver:** EPC, grid integration, O&M, regulatory navigation.

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## WHY THIS ALLIANCE WINS

**Kader builds it. AVESTA powers it. Giza connects it.**

No foreign EPC. No Chinese middlemen. No currency risk. No six-month spare part waits.

One contract. One team. One throat to choke.

From raw steel to grid-connected electrons - all inside Egypt.

AVESTA KADER

Powered by Avesta Belgium

# Building the Bright Future Together

*The Grid of Tomorrow Needs Investment Today.*

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